

Supercritical extraction of carotenoids and lipids from buriti (*Mauritia flexuosa*), a fruit from the Amazon region

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Received 9 February 1998; received in revised form 27 August 1998; accepted 31 August 1998

Abstract

Pulp of the buriti fruit (*Mauritia flexuosa*, Mart.) was extracted with supercritical CO₂ to obtain oil fractions with a high concentration of vitamins, especially β -carotene. The raw material consisted of a mixture of pulp and peel that was scraped off the immature fruit. The extracted oil was analyzed by gas chromatography and spectrophotometry, and results were compared with those obtained by extraction using hexane as solvent. The oil extracted with hexane showed a content of about 1% carotene. Extraction with supercritical CO₂ was capable of removing about 80% of the initial carotene content. Extraction experiments were carried out at pressures of 20 and 30 MPa and temperatures of 313 and 328 K. The extraction curves typically showed the three regions found in supercritical fluid extraction of natural products, i.e. beginning with a straight line characteristic of the constant rate extraction period, followed by the falling rate period, and ending with the diffusion rate period. Experimental data were correlated using the model proposed by Sovová. The model showed good agreement with experimental data. The mass transfer coefficients of the solid phase were obtained by adjusting the model equation. The solubility values calculated are within the range for common vegetable oils. © 1999 Elsevier Science B.V. All rights reserved.

Keywords: Carotene; Extraction; *Mauritia flexuosa*; Natural products; Solubility; Supercritical fluids

1. Introduction

Fats and oils play an important role in the food industry and are important in human nutrition for a variety of reasons. They are the main source of energy for the human body compared with their

weight. Whereas 1 g of carbohydrate provides about 16.7 kJ of energy, 1 g of fat provides as much as 37.7 kJ [1]. Nevertheless, their wide use in the food industry is due to their ability to dissolve various flavor and aroma compounds, and a distinct consistency to various food products. Nutritional considerations have influenced the use of fats and oils in recent years. The consumption of large amounts of saturated fatty acids and

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cholesterol-containing animal fat is regarded as a major risk factor to arterial diseases [2]. Moreover, oils, particularly vegetable oils, are becoming increasingly important as suppliers of fat-soluble vitamins such as vitamins A, D, E, and K. They are also an important source of essential unsaturated fatty acids, which cannot be synthesized by the human body. In order to meet nutritional requirements, new vegetable oil resources are being sought as a source of vitamins and essential fatty acids.

The palm fruits from the Amazon region promise to be an alternative and abundant source of vegetable oils with high nutritional value [3]. Currently, only a few of these palms are being cultivated and exploited, and the processing of many fruits is still carried out on a commercially small scale. In general, knowledge of the processing of palm fruits is the heritage of native population rather than a widely used technique. Therefore, it is very important to investigate the possibility of using some of the palm fruits as a source of vegetable oils and vitamins.

Since development of the process for removing caffeine from coffee [4] and the extraction of the essence of hop [5], the extraction of natural products with carbon dioxide has received much attention. There are many examples in the literature of extraction of oily substances from different raw materials such as canola seeds [6], bacuri shell and seed [7,8], tucumã seed [9], passion fruit [8,10], as well as herbaceous and spice materials [11–13], Turkish lavender flowers [14], and fungal material [15]. Lately, there has been great interest in supercritical fluid extraction of pigments such as the carotene from different raw materials such as alfalfa [16], sweet potatoes [17], microalgae [18], and the residue of the mechanical processing of palm oil [19].

The objective of this work was to explore the potential application of supercritical CO₂ to the extraction of carotene from buriti (*Mauritia flexuosa*), an Amazon-region palm fruit. The effect of operational conditions on the composition of the extracts was investigated. An extraction model was used to describe the mass transfer aspects of the process.

2. Material and methods

The buriti fruit (Fig. 1) was collected from a palm tree located in the fluvial region of Abaetetuba (Pará, Brazil). Its average composition was 20% shell and pulp, 30% white cellulose layer, and 50% seed (wt/wt of fruit). The pulp plus the shell of the fresh fruit have an average moisture content of 30% as determined using an oven with air circulation (model 179, Fabbe, São Paulo, Brazil). White Martins (Belém, Brazil) provided carbon dioxide 99.99% pure.

2.1. Raw material preparation and characteristics

The freshly picked fruit was rather hard and immature. It was stored in room conditions ($\sim 32^{\circ}\text{C}$ and $\sim 80\%$ relative humidity) for 5 to 7 days in order to ripen. The shell and pulp were used in the experiments. Chips of fruit (shell and pulp) were removed up to the white cellulose layer (Fig. 1). The solid material was dried in an oven with air circulation (model 179, Fabbe, São Paulo, Brazil) at 333 K for 24 h. Afterwards, it was ground in a comminuting mill (model A-70, Gerätetechnik, Germany). The dried material was packed in plastic bags and kept in a refrigerator at 278–283 K. The dried solid material was sifted for 30 min in a sieve shaker (Granutest, São Paulo, Brazil) fitted with sieves of different sizes. The dried shell and pulp of the buriti fruit had the following characteristics: 11.0 ± 0.5 wt% moisture;

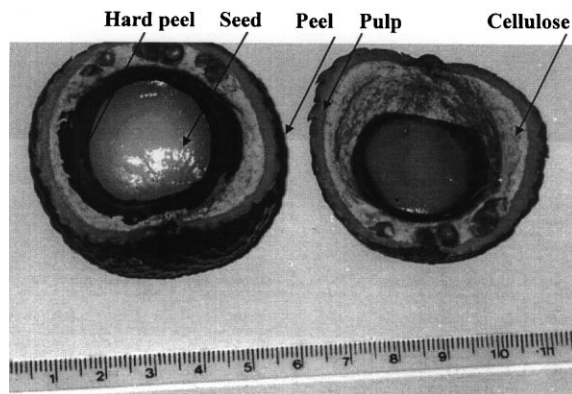


Fig. 1. Photograph of the raw material (buriti fruit) used in the experiments.

7.5 ± 0.1 wt% lipids as measured by the AOAC 920.39 method [20]; 5.0 ± 0.5 wt% proteins as measured by the AOAC 320.176 method [20].

2.2. Experimental apparatus

The experiments were carried out in the supercritical extraction unit available in the Department of Chemical Engineering at the Federal University of Pará (Brazil), as part of a joint cooperative project with TUHH (Technische Universität Hamburg–Harburg). A schematic diagram of the extraction unit is shown in Fig. 2. It consists of one jacketed autoclave (Metalwerkstatt, TUHH), a diaphragm-type compressor (model MKZ 120-50, Andreas Hofer, Mülheim, Germany), and a control unit (Elektrowerkstatt, TUHH). The extractor has a volume of 1000 cm^3 (6 cm internal diameter and 36 cm height) and the separator 130 cm^3 . A thermostatic bath (model N3, Haake Mess-Technik GmbH, Karlsruhe, Germany) was used to maintain the operating temperature constant inside the jacketed autoclave. The temperature inside the extractor was measured with an NiCr–Ni thermocouple (model MTE-303, SAB GmbH, Viersen, Germany) with a precision of 2.5 K in the graphic register. Pressures inside the extractor and separator were measured by means

of a gauge manometer (0–40 MPa, model 332.30, Wika Alexander Wiegand GmbH, Klingenberg, Germany) with a precision of 0.5 MPa. A micrometering valve (model SW 2081, Autoclave Engineers, USA) was used to control the solvent mass flow rate. The mass flow rate of carbon dioxide was measured using a flow totalizer (Bopp and Reuter Mess-Technik GmbH, Maisach, Germany).

2.3. Extraction procedure

Carbon dioxide was delivered at the required pressure by the membrane compressor and passed through a porous plate in order to assure a homogeneous flow of carbon dioxide through the fixed bed of solid material placed inside the extractor. After passing through the micrometering valve (control valve), the supercritical carbon dioxide containing the solubilized oil expanded in the small stainless steel tube used as separator and the extract was collected in a glass tube placed inside it. The samples were collected every 20 min for the first 2 h and every 30 min afterwards. The expanded carbon dioxide then passed through a flow meter and was released to the atmosphere. The pressure and temperature inside the extraction vessel varied within the experimental error.

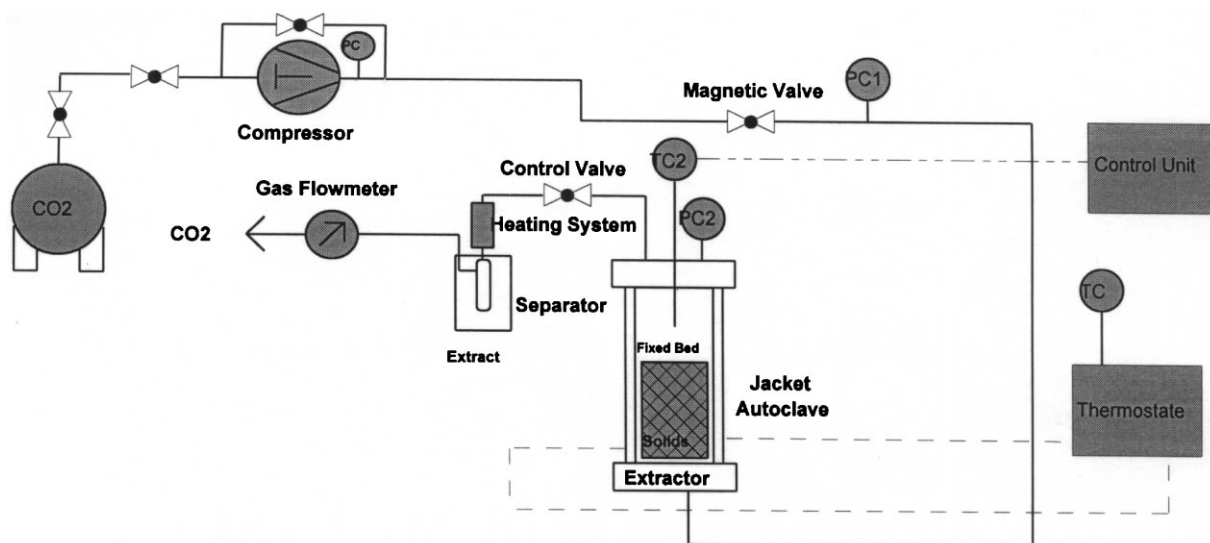


Fig. 2. Flowsheet of the supercritical extraction unit.

2.4. Analysis of extracts

The extracts were analyzed with respect to total carotenes in a spectrophotometer and to other components (free fatty acids and tocopherols) by gas chromatography.

2.4.1. Gas chromatographic analysis

The extracts (20–30 mg) were diluted in 1 ml of a standard solution (0.9 mg ml^{-1} squalane) prepared with acetone P.A. (>99%, Merck, Darmstadt, Germany), *n*-hexane P.A. (>99%, Merck, Darmstadt, Germany) in the proportion (1:1 vol./vol.), and squalane (>99%, Riedel de Haën, internal standard). The fatty acid composition was analyzed using a gas chromatograph (model 5890 A, Hewlett Packard) equipped with FID detector and a fused-silica column DB-5 ($30 \text{ m} \times 0.25 \text{ mm} \times 0.1 \text{ } \mu\text{m}$, J&W Scientific, Folsom, CA, USA). The carrier gas was nitrogen ($2 \text{ } \mu\text{l min}^{-1}$). The sample size injected was $1 \text{ } \mu\text{L}$. The temperature was kept at 343 K for 2 min, then raised to 523 K at 25 K min^{-1} and to 623 K at 4 K min^{-1} , and kept at this temperature for 15 min. The detector temperature was 633 K and that of the injector was 593 K. The chromatographic response factors for the different free fatty acids and tocopherols were determined by analyzing standard solutions prepared gravimetrically. Palmitic acid (>99%, Merck, Darmstadt, Germany), oleic acid (>99%, Merck, Darmstadt, Germany), and linoleic acid (>98%, Merck, Darmstadt, Germany) of an analytical grade were used to prepare the standard solutions. The chromatograms show main component retention times (except for carotenes) for oil extracted with supercritical carbon dioxide. This methodology is similar to that used by Stoldt [21], except for the fact that nitrogen instead of helium was used as carrier gas.

2.4.2. Spectrophotometric analysis

Extracts (20–30 mg) were analyzed using a spectrophotometer (model UV-120-020, Shimadzu, Tokyo, Japan). They were diluted in 25 or 50 ml of a 7:3 vol./vol. solvent mixture of *n*-hexane and acetone P.A. (>99%, Merck, Darmstadt, Germany) and absorbance was read at 450 nm.

Carotenes in the extracts were calculated in terms of β -carotene, using a standard absorbance curve calibrated with β -carotene (>99%, Merck, Darmstadt, Germany). The standard curve was prepared with 0.1 g of β -carotene diluted up to 100 ml with the 7:3 vol./vol. solvent mixture of *n*-hexane and acetone. Aliquots were taken from this solution and diluted to five different concentrations. Absorbance was read at 450 nm.

3. Results and discussion

Table 1 shows the composition of the oil extracted with supercritical CO_2 ($P=20 \text{ MPa}$, $T=313 \text{ K}$). A Soxhlet extraction with hexane using 16 g of solid material was also performed. The extract obtained had the following composition: $7.5 \pm 0.1 \text{ wt\%}$ oil and 77.7 mg of β -carotene per 100 g of solid material.

No information was found in the literature concerning the extraction of lipids from buriti shell and pulp. Therefore, a preliminary set of experiments was performed to evaluate the fixed bed characteristics (particle size distribution and amount of dried solid material required to form the bed) and solvent flow rate. Based on the results, the size of the solid material chosen showed the following distribution: 1.5% had an average diameter of 2.38 mm; 14.8% of 1.19 mm; 41.4% of 0.595 mm; 14.3% of 0.355 mm; 24.8% of 0.21 mm; 2.6% smaller than 0.21 mm. This size distribution required a bed formed with $100.0 \pm 0.5 \text{ g}$ of dried buriti which resulted in a bed with a height of 6 cm. The fixed bed characteristics were: a true

Table 1
Composition of the oil from the shell and pulp of the buriti extracted with supercritical CO_2 at 20 MPa and 313 K

Substance	Amount
Carotenoids (ppm)	$(1043 \pm 8) \times 10^1$
Tocopherols (ppm)	$(19.6 \pm 1) \times 10^3$
Oleic acid value ($\text{g}_{\text{oleic acid}}/\text{100 g}_{\text{oil}}$)	10.8 ± 0.7
Free fatty acid composition (%)	
palmitic acid (C-16:0)	17.34
oleic acid (C-18:1)	78.73
linoleic acid (C-18:2)	3.93

density of $0.88 \pm 0.03 \text{ g cm}^{-3}$, as measured by the sand pycnometry technique [22] after extraction; an apparent density of $0.59 \pm 0.03 \text{ g cm}^{-3}$; an average particle diameter of $0.66 \pm 0.02 \text{ mm}$, calculated as suggested in the ASAES-319.2 method [23]; a porosity of 0.33.

3.1. The effects of temperature, pressure, and solvent flow rate on extraction yield

A factorial experiment with replication for temperatures of 313 and 328 K and pressures of 20 and 30 MPa was performed to study the effect of pressure and temperature on extraction yield. To verify the effect of flow rate on extraction, an experiment with replication was also conducted at 30 MPa, 313 K, and an average flow rate of 25.8 g min^{-1} of CO_2 .

Fig. 3 shows the influence of operating pressure on mass transfer kinetics for the extraction carried out at 313 K and two different operating pressures (20 and 30 MPa). In both cases the extraction curves show the three distinct regions usually found in the extraction of natural products using supercritical fluids: constant extraction rate period, falling rate period (transition) and diffusion-controlled rate period. The results show that the slope of the straight line representing the constant extraction rate period increases as the pressure increases. This effect is due to the increase in solubility of the solute mixture in the gaseous phase. As pressure increases, all easily accessible oil at the surface

of the particles is dissolved rapidly in the fluid phase because of an increase in the solvation power of carbon dioxide. Solubility in the supercritical phase provides information on the amount of solvent, and the energy required recycling the solvent necessary to extract oil from the solid matrix. In cases where the target component to be extracted (e.g. fat-soluble components) is the easily accessible oil at the surface of the particles, there would be no reason to extend extraction up to the end, because of the drastic increase in the solvent compression cost of increasing the yield. Unfortunately, this is not the case with carotene extraction, since extract fractions with high carotene contents are only obtained in the diffusion-controlled extraction period, as will be discussed later.

Using the experimental data, the mass transfer rate M_{CER} and the concentration of solute in the supercritical phase Y_{CER} at the bed outlet for the constant rate period were calculated. The procedure employed was that explained by Ferreira et al. [24]. The duration of the constant extraction rate period t_{CER} was also computed. Table 2 shows the results for all the experiments. The analysis of variance performed showed that the effects of solvent flow rate, pressure, and temperature on mass transfer rate M_{CER} are significant and that the effects of solvent flow rate and pressure are more significant ($p=10^{-4}$) than that of temperature ($p=0.0132$). The duration of the constant extraction rate period t_{CER} was significantly affected by solvent flow rate ($p=10^{-4}$), pressure ($p=10^{-4}$), and temperature ($p=0.0394$). Total yield was also significantly affected by the same variables ($p=0.0042$; $p=0.0003$; $p=0.0017$ respectively), as well as by the interaction between temperature and pressure ($p=0.0043$). On the other hand, for concentration of solute in the supercritical phase at the bed outlet Y_{CER} , the effects of temperature ($p=0.0021$), pressure ($p=10^{-4}$), and solvent flow rate ($p=10^{-4}$) were significant. The velocity of the supercritical CO_2 in the extractor is low (Table 2); it is rather likely that equilibrium solubility of the oil at the respective conditions was reached. In order to determine, whether the Y_{CER} indicates solubility (Y^*) the response variable Y_{CER} was fitted as a function of temperature and

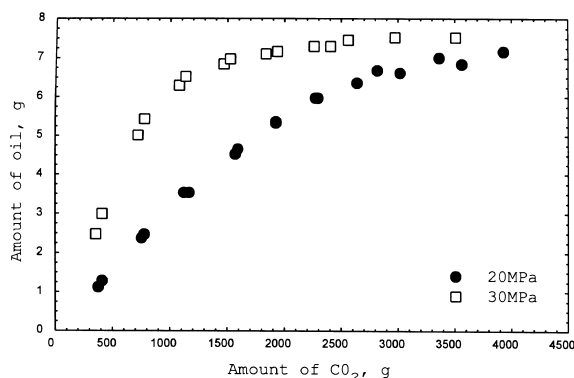


Fig. 3. Extraction of buriti at 313 K, solvent flow rate of 0.307 g s^{-1} , and pressures of 20 and 30 MPa.

Table 2
Experimental conditions and results

T (K)	Q (g s ⁻¹ CO ₂)	M_{CER} 10 ⁴ (g s ⁻¹ oil)	Y_{CER} 10 ⁴ (g g ⁻¹ CO ₂)	t_{CER} (min)	Yield _{CER} (mass%) ^a	t_{TOTAL} (min)	Yield (mass%) ^a
20 MPa							
313	0.31 ± 0.02	8.8 ± 0.3	28 ± 1	119.2	6.3	210	7.2
313	0.30 ± 0.03	8.8 ± 0.3	29 ± 2	111.7	5.9	195	6.9
328	0.28 ± 0.01	4.8 ± 0.1	17 ^b	139.8	4.1	200	5.4
328	0.29 ± 0.06	4.5 ± 0.3	15 ± 1	149.0	4.0	195	5.1
328	0.30 ± 0.02	5.3 ± 0.1	18 ^b	124.6	3.9	140	4.7
30 MPa							
313	0.31 ± 0.01	21 ± 1	68 ± 4	53.6	6.8	190	7.5
313	0.31 ± 0.02	19 ± 2	65 ± 8	55.8	6.6	160	7.5
328	0.30 ± 0.03	18 ± 2	60 ± 7	60.0	6.5	160	7.6
328	0.28 ± 0.01	17 ± 2	61 ± 7	62.0	6.3	140	7.3
313	0.43 ± 0.03	26 ± 1	62 ± 3	36.2	5.7	95	7.1
313	0.43 ± 0.04	31 ± 2	70 ± 4	36.7	6.7	95	7.8

^a Mass of extract/mass of dried material.

^b Standard deviation is less than 1×10^4 .

pressure with the solvent flow rate as a covariate. When the effects of temperature and pressure were removed, the effect of the solvent flow rate was not significant ($p=0.6084$). This result confirms that the experiments were conducted at the limit where equilibrium between the oil and the solvent was reached, and the solvent left the extractor saturated with oil. Results similar to those discussed above have been observed for the extraction of black pepper essential oil [25]. Fig. 4 shows the extraction curves at 30 MPa and 313 K, and solvent mass flow rates of 0.307 g s⁻¹ and

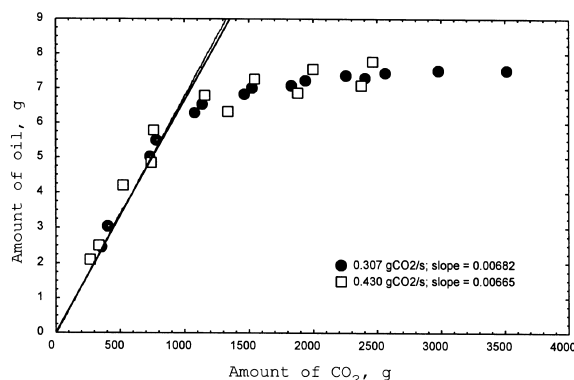


Fig. 4. Curve of extraction of buriti at 30 MPa and 313 K with different mass flows.

0.430 g s⁻¹. Table 3 presents the mean value of solubility determined at each experimental condition. Using the equation of del Valle and Aguilera [26] the solubilities at the various conditions were calculated; these values are also shown in Table 3. Experimental and calculated solubilities showed at 313 K a deviation below 20% and at 328 K larger than 33%. These differences may be explained considering the differences in composition of the extracts. The authors developed the solubility equation considering solubility data for oil of soybean, canola, etc. The composition of these extracts differs from that of the oil obtained from the shell and pulp of buriti. Although the effect of temperature on the composition was not determined for the buriti extract, based on our previous results for other seed oils [8,10], we assume that it can vary with temperature as well with pressure. However, further investigation is required to elucidate the mechanisms of interaction between supercritical CO₂ and the soluble material from the buriti shell and pulp.

3.2. The carotene concentration

The course of carotene extraction was observed by measuring the concentration in each fraction

Table 3

Mass transfer coefficients and solubilities obtained by fitting experimental data to model equation

<i>P</i> (MPa)	<i>T</i> (K)	CO ₂ flow (g s ⁻¹)	<i>k_sa₀</i> 10 ⁴ (s ⁻¹)	<i>Y</i> [*] 10 ³ (g g ⁻¹ CO ₂)	<i>Y</i> [*] 10 ³ (g g ⁻¹ CO ₂) ^a	Deviation (%)
20	313	0.31	6.5	2.86	3.44	16.9
30	313	0.31	0.65	6.82	7.49	9.0
30	313	0.43	2.95	6.65	7.49	11.2
20	328	0.29	— ^b	1.74	2.56	31.9
30	328	0.29	1.52	6.05	8.08	25.2

^a Calculated with del Valle and Aguilera equation [27].^b The diffusion-controlled extraction rate period was not reached.

collected at regular intervals of time. The results are shown in Fig. 5. During extraction, carotene concentration increases as the amount of CO₂ increases. Comparing Figs. 3 and 5 leads to the conclusion that the largest amount of carotenes was extracted in the diffusion-controlled period. Therefore, one may expect carotenes to be found in the oleaginous fractions located deep inside the solid particles. At the beginning of extraction, lipids in or near the solid surface are quickly dissolved in the CO₂; then, when these begin draining, the carotenes now dissolved in less oil are extracted by the CO₂, resulting in a product with a higher carotene concentration. After about 4000 g of CO₂ had been used, more than 50 mg of carotene had been extracted, as shown in Fig. 5. The extraction with hexane, where about 77 mg (or 10 000 ppm) of carotene was extracted, indicates that about 80% of the total carotenes was

extracted with CO₂; of this total, however, approximately 50% was obtained in oil fractions with a high carotene concentration.

3.3. Extraction model

Of the several mathematical models for supercritical extraction of oils from a solid matrix, the one developed by Sovová [27] was selected. Oil leaving the solid phase is transported into the fluid phase. It is assumed that oil on the surface of the material is dissolved rapidly. First, the oil begins to deplete on the bottom of the fixed bed, and afterwards the oil is transported out of the solid material. This layer moves upwards until all easily accessible oil is dissolved; the main mass transfer resistance is due to diffusion in the solid material.

The equations governing the mass transfer process are obtained by performing a material balance in both solid and fluid phases. The model equations have the following form:

$$-\rho_s(1-\epsilon)\frac{\partial X}{\partial t}=J(X,Y) \quad \text{solid phase} \quad (1)$$

$$\rho_f U \frac{\partial Y}{\partial h}=J(X,Y) \quad \text{fluid phase} \quad (2)$$

where ρ_s and ρ_f are the densities of the solid and fluid phases; U is the superficial velocity; ϵ is the bed porosity; X and Y are the mass ratios of the solute to the inert solid and the solvent respectively; t is time and h is the axial distance of the bed.

The term $J(X,Y)$ must be replaced by an adequate expression for the mass transfer resistance from one phase into the other. Three expressions

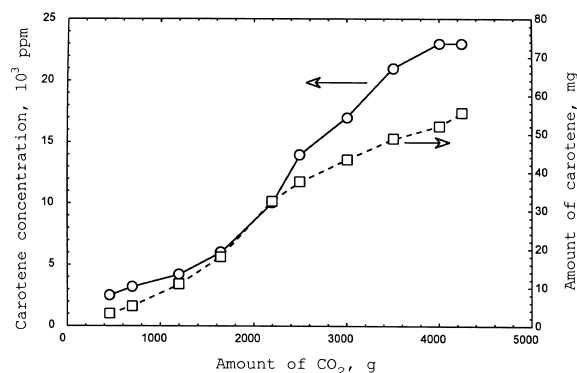


Fig. 5. The course of carotene extraction at 20 MPa and 313 K. Carotene concentration in the extract and amount of carotene collected as a function of the amount of carbon dioxide used.

are obtained which describe the constant extraction rate period, the diffusion-controlled extraction rate period, and the falling extraction rate period in which the layer of depletion of easily accessible oil moves from bottom to top. The shape of the extraction curves (Fig. 3) allows one to neglect the falling rate period. Therefore, only the constant and the diffusion-controlled rate periods will be considered further.

The mass of oil extracted m_{ext} for the constant extraction rate period is

$$m_{\text{ext}} = NqY^*[1 - \exp(-Z)] \quad t \leq t_{\text{CER}} \quad (3)$$

where N is the mass of inert solid, q is the mass ratio of solvent to inert solid at time t , Y^* is the solubility and the parameter Z is given by

$$Z = k_f a_0 \rho_f / [q^* (1 - \epsilon) \rho_s] \quad (4)$$

where $k_f a_0$ is the fluid-phase volumetric mass transfer coefficient and q^* is the average mass ratio of solvent to inert solid ($= Q/N$). The expression for the diffusion-controlled extraction rate period has the following form:

$$m_{\text{ext}} = N(X_0 - [Y^*/W] \ln\{1 + [\exp(WX_0/Y^*) - 1] \times \exp[W(q_{\text{CER}} - q)](X_0 - X_k)/X_0\}) \quad t > t_{\text{CER}} \quad (5)$$

where q_{CER} is the mass ratio of solvent to inert solid at t_{CER} ; X_0 and X_k are the mass ratios of initial and easily accessible oil respectively. The parameter W is given by

$$W = k_s a_0 / [q^* (1 - \epsilon)] \quad (6)$$

where $k_s a_0$ is the volumetric mass transfer coefficient of the solid phase.

The falling rate period, in which the easily accessible oil is already depleted at the bottom of the fixed bed but not at the top, had been neglected. Therefore, the beginning of the diffusion-controlled extraction rate period was taken as the end of the constant extraction rate period and was calculated using the procedure employed by Ferreira et al. [24].

Once, as discussed earlier, the constant rate period coincides with the saturation or equilibrium period, Eq. (3) can be simplified and the first part of the extraction curves can be described by

$$m_{\text{ext}} = NqY^* \quad (7)$$

About 85% of the oil was extracted in the constant extraction rate period. The total amount of extractable oil was assumed to be 7.5%, based on the results of the Soxhlet extraction. The total sample mass was 100 g; thus, the value for N was 92.5 g. The initial mass ratio of oil X_0 was 0.0811, and that of easily accessible oil X_k was 0.0689. These parameters were taken to be valid for all experimental conditions employed. Solubility and values of the parameter $k_s a_0$ obtained by fitting the model to the experimental data for the diffusion-controlled extraction rate period are listed in Table 3. The constant extraction rate period was longer at 20 MPa than at 30 MPa. Y_{CER} and the total yield at both pressures were comparable. On the other hand, the mass transfer rates M_{CER} were about seven times larger at 30 MPa. This explains the higher value of $k_s a_0$ obtained at 20 MPa and 313 K. At 20 MPa and 328 K t_{TOTAL} is about $1.2t_{\text{CER}}$, indicating that the diffusion-controlled step can be neglected at these conditions. According to the model this was also the case, once no value for the parameter $k_s a_0$ was obtained.

Fig. 6 compares experimental data with data calculated using the model proposed by Sovová [27]. As can be observed, agreement between experimental and calculated values is very good. Therefore, the model chosen, along with the simplifying assumptions, adequately described the supercritical extraction of lipids from buriti shell and pulp.

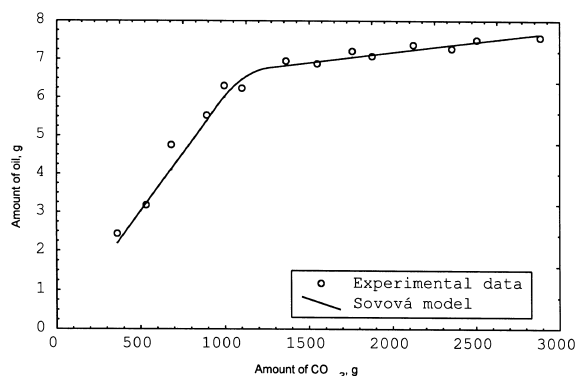


Fig. 6. Extraction curve for buriti at 30 MPa and 328 K.

4. Conclusions

The experimental results show that the extraction curves have the three distinct regions usually found in the extraction of natural products using supercritical fluids: constant extraction rate period, falling rate period (transition) and diffusion-controlled rate period. The statistical analysis performed showed that the effects of solvent flow rate, pressure, and temperature on mass transfer rate M_{CER} , the constant extraction rate period t_{CER} , and the total yield were significant. The effect of the solvent flow rate on the concentration of solute in the supercritical phase Y_{CER} at the bed outlet was not significant. During extraction, carotene concentration increased as the amount of CO_2 increased; therefore, large amounts of carotenes were removed, in the decreasing or diffusion-controlled extraction rate period. The agreement between experimental and values calculated using the mathematical model proposed was very good.

Acknowledgments

The authors want to express their gratitude to Dr Ademir Petenate (IMECC, Unicamp, Brazil) for his assistance with the statistical analysis. This research was partially financed by CNPq–Brazil. White Martins provided the carbon dioxide. TUHH provided the scholarship for Georg Reber's 'diplomarbeit' studies.

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